Experiment No:14
Experiment to understand feature scaling.

AIM:
To understand feature scaling.

DESCRIPTION:
Understand the importance of feature scaling

CODE:

```
import numpy as np
import pandas as pd

df = pd.read_csv('/content/50_Startups.csv')

print(df)

df["State"] = df["State"].fillna(df["State"].mode()[0])

features = df.iloc[:, :-1].values
label = df.iloc[:, -1].values

from sklearn.impute import SimpleImputer

age = SimpleImputer(strategy="mean", missing_values=np.nan)
salary = SimpleImputer(strategy="mean", missing_values=np.nan)

age.fit(features[:, [0]])
salary.fit(features[:, [1]])

features[:, [0]] = age.transform(features[:, [0]])
features[:, [1]] = salary.transform(features[:, [1]])

print(features)
```

```
from sklearn.preprocessing import OneHotEncoder

oh = OneHotEncoder(sparse_output=False)
State = oh.fit_transform(features[:, [3]])

print(State)

final_set = np.concatenate((State, features[:, [0, 1, 2]]),
axis=1)

print(final_set)

from sklearn.preprocessing import StandardScaler

sc = StandardScaler()
sc.fit(final_set)

feat_standard_scaler = sc.transform(final_set)

print(feat_standard_scaler)

from sklearn.preprocessing import MinMaxScaler

mms = MinMaxScaler(feature_range=(0, 1))
mms.fit(final_set)

feat_minmax_scaler = mms.transform(final_set)

print(feat_minmax_scaler)
```

OUTPUT:

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83

1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94
5	131876.90	99814.71	362861.36	New York	156991.12
6	134615.46	147198.87	127716.82	California	156122.51
7	130298.13	145530.06	323876.68	Florida	155752.60
8	120542.52	148718.95	311613.29	New York	152211.77
9	123334.88	108679.17	304981.62	California	149759.96
10	101913.08	110594.11	229160.95	Florida	146121.95
11	100671.96	91790.61	249744.55	California	144259.40
12	93863.75	127320.38	249839.44	Florida	141585.52
13	91992.39	135495.07	252664.93	California	134307.35
14	119943.24	156547.42	256512.92	Florida	132602.65
15	114523.61	122616.84	261776.23	New York	129917.04
16	78013.11	121597.55	264346.06	California	126992.93
17	94657.16	145077.58	282574.31	New York	125370.37
18	91749.16	114175.79	294919.57	Florida	124266.90
19	86419.70	153514.11	0.00	New York	122776.86
20	76253.86	113867.30	298664.47	California	118474.03
21	78389.47	153773.43	299737.29	New York	111313.02
22	73994.56	122782.75	303319.26	Florida	110352.25
23	67532.53	105751.03	304768.73	Florida	108733.99
24	77044.01	99281.34	140574.81	New York	108552.04
25	64664.71	139553.16	137962.62	California	107404.34
26	75328.87	144135.98	134050.07	Florida	105733.54
27	72107.60	127864.55	353183.81	New York	105008.31
28	66051.52	182645.56	118148.20	Florida	103282.38
29	65605.48	153032.06	107138.38	New York	101004.64
30	61994.48	115641.28	91131.24	Florida	99937.59
31	61136.38	152701.92	88218.23	New York	97483.56
32	63408.86	129219.61	46085.25	California	97427.84
33	55493.95	103057.49	214634.81	Florida	96778.92
34	46426.07	157693.92	210797.67	California	96712.80
35	46014.02	85047.44	205517.64	New York	96479.51
36	28663.76	127056.21	201126.82	Florida	90708.19
37	44069.95	51283.14	197029.42	California	89949.14

```
38   20229.59       65947.93        185265.10    New York   81229.06
39   38558.51       82982.09        174999.30  California   81005.76
40   28754.33      118546.05        172795.67  California   78239.91
41   27892.92       84710.77        164470.71     Florida   77798.83
42   23640.93       96189.63        148001.11  California   71498.49
43   15505.73      127382.30         35534.17    New York   69758.98
44   22177.74      154806.14         28334.72  California   65200.33
45    1000.23      124153.04          1903.93    New York   64926.08
46    1315.46      115816.21        297114.46     Florida   49490.75
47       0.00      135426.92             0.00  California   42559.73
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49       0.00      116983.80         45173.06  California   14681.40
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 [1.         0.         0.         0.46116861 0.47642362 0.63305328]
 [0.         0.         1.         0.47408436 0.78021012 0.63532724]
 [0.         1.         0.         0.4475048  0.54429273 0.64291963]
 [0.         1.         0.         0.40842369 0.4146383  0.64599195]
 [0.         0.         1.         0.46594728 0.3653876  0.29796428]
 [1.         0.         0.         0.39107967 0.67195793 0.29242745]
 [0.         1.         0.         0.45557444 0.70684477 0.28413435]
 [0.         0.         1.         0.43609283 0.58297807 0.74861321]
 [0.         1.         0.         0.39946683 1.         0.25042853]
 [0.         0.         1.         0.39676926 0.77456642 0.22709197]
 [0.         1.         0.         0.37493063 0.48992809 0.19316302]
 [0.         0.         1.         0.36974101 0.77205322 0.18698856]
 [1.         0.         0.         0.38348453 0.5932935  0.09768292]
 [0.         1.         0.         0.33561668 0.39413365 0.45494286]
```

```
[1.         0.         0.         0.2807759  0.81005496 0.44680961]
[0.         0.         1.         0.2782839  0.25703165 0.43561799]
[0.         1.         0.         0.17335288 0.57682456 0.42631115]
[1.         0.         0.         0.26652654 0.         0.41762624]
[0.         0.         1.         0.12234465 0.11163611 0.39269043]
[1.         0.         0.         0.23319442 0.24130912 0.3709309 ]
[1.         0.         0.         0.17390063 0.51204073 0.36626005]
[0.         1.         0.         0.16869099 0.25446874 0.34861436]
[1.         0.         0.         0.14297577 0.34185188 0.31370517]
[0.         0.         1.         0.09377566 0.57930693 0.07531871]
[1.         0.         0.         0.13412668 0.78807166 0.06005866]
[0.         0.         1.         0.0060492  0.5547241  0.0040356 ]
[0.         1.         0.         0.00795565 0.49125975 0.62976785]
[1.         0.         0.         0.         0.64054682 0.        ]
[0.         0.         1.         0.00327821 0.00350184 0.        ]
[1.         0.         0.         0.         0.50014806 0.09574943]]
```

RESULT:

Thus, feature scaling was successfully implemented using the StandardScaler and MinMaxScaler methods. The categorical State feature was converted into numerical values using OneHotEncoder, and missing values were handled using SimpleImputer. The experiment demonstrated how feature scaling transforms data into a suitable format for machine learning applications.